

Early atomic physics



1.1 Introduction

The origins of atomic physics were entwined with the development of quantum mechanics itself ever since the first model of the hydrogen atom by Bohr. This introductory chapter surveys some of the early ideas, including Einstein’s treatment of the interaction of atoms with radiation, and a classical treatment of the Zeeman effect. These methods, developed before the advent of the Schrödinger equation, remain useful as an intuitive way of thinking about atomic structure and transitions between the energy levels. The ‘proper’ description in terms of atomic wavefunctions is presented in subsequent chapters.

Before describing the theory of an atom with one electron, some experimental facts are presented. This ordering of experiment followed by explanation reflects the author’s opinion that atomic physics should not be presented as applied quantum mechanics, but it should be motivated by the desire to understand experiments. This represents what really happens in research where most advances come about through the interplay of theory and experiment.

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1.2 Spectrum of atomic hydrogen

It has long been known that the spectrum of light emitted by an element is characteristic of that element, e.g. sodium in a street lamp, or burning in a flame, produces a distinctive yellow light. This crude form of spectroscopy, in which the colour is seen by eye, formed the basis for a simple chemical analysis. A more sophisticated approach using a prism, or diffraction grating, to disperse the light inside a spectrograph shows that the characteristic spectrum for atoms is composed of discrete lines that are the ‘fingerprint’ of the element. As early as the 1880s, Fraunhofer used a spectrograph to measure the wavelength of lines, that had not been seen before, in light from the sun and he deduced the existence of a new element called helium. In contrast to atoms, the spectra of molecules (even the simplest diatomic ones) contain many closely-spaced lines that form characteristic molecular bands; large molecules, and solids, usually have nearly continuous spectra with few sharp features. In 1888, the Swedish professor J. Rydberg found that the spectral

lines in hydrogen obey the following mathematical formula:

$$\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{n'^2} \right), \quad (1.1)$$

where n and n' are whole numbers; R is a constant that has become known as the Rydberg constant. The series of spectral lines for which $n = 2$ and $n' = 3, 4, \dots$ is now called the Balmer series and lies in the visible region of the spectrum.¹ The first line at 656 nm is called the Balmer- α (or H_α) line and it gives rise to the distinctive red colour of a hydrogen discharge—a healthy red glow indicates that most of the molecules of H_2 have been dissociated into atoms by being bombarded by electrons in the discharge. The next line in the series is the Balmer- β line at 486 nm in the blue and subsequent lines at shorter wavelengths tend to a limit in the violet region.² To describe such series of lines it is convenient to define the reciprocal of the transition wavelength as the wavenumber $\tilde{\nu}$ that has units of m^{-1} (or often cm^{-1}),

$$\tilde{\nu} = \frac{1}{\lambda}. \quad (1.2)$$

Wavenumbers may seem rather old-fashioned but they are very useful in atomic physics since they are easily evaluated from measured wavelengths without any conversion factor. In practice, the units used for a given quantity are related to the method used to measure it, e.g. spectroscopes and spectrographs are calibrated in terms of wavelength.³ A photon with wavenumber $\tilde{\nu}$ has energy $E = hc\tilde{\nu}$. The Balmer formula implicitly contains a more general empirical law called the Ritz combination principle that states: the wavenumbers of certain lines in the spectrum can be expressed as sums (or differences) of other lines: $\tilde{\nu}_3 = \tilde{\nu}_1 \pm \tilde{\nu}_2$, e.g. the wavenumber of the Balmer- β line ($n = 2$ to $n' = 4$) is the sum of that for Balmer- α ($n = 2$ to $n' = 3$) and the first line in the Paschen series ($n = 3$ to $n' = 4$). Nowadays this seems obvious since we know about the underlying energy-level structure of atoms but it is still a useful principle for analyzing spectra. Examination of the sums and differences of the wavenumbers of transitions gives clues that enable the underlying structure to be deduced, rather like a crossword puzzle—some examples of this are given in later chapters. The observed spectral lines in hydrogen can all be expressed as differences between energy levels, as shown in Fig. 1.1, where the energies are proportional to $1/n^2$. Other series predicted by eqn 1.1 were more difficult to observe experimentally than the Balmer series. The transitions to $n = 1$ give the Lyman series in the vacuum ultraviolet region of the spectrum.⁴ The series of lines with wavelengths longer than the Balmer series lie in the infra-red region (not visible to the human eye, nor readily detected by photographic film—the main methods available to the early spectroscopists). The following section looks at how these spectra can be explained theoretically.

¹The Swiss mathematician Johann Balmer wrote down an expression which was a particular case of eqn 1.1 with $n = 2$, a few years before Johannes (commonly called Janne) Rydberg found the general formula that predicted other series.

²A spectrum of the Balmer series of lines is on the cover of this book.

³In this book transitions are also specified in terms of their frequency (denoted by f so that $f = c\tilde{\nu}$), or in electron volts (eV) where appropriate.

⁴Air absorbs radiation at wavelengths shorter than about 200 nm and so spectrographs must be evacuated, as well as being made with special optics.

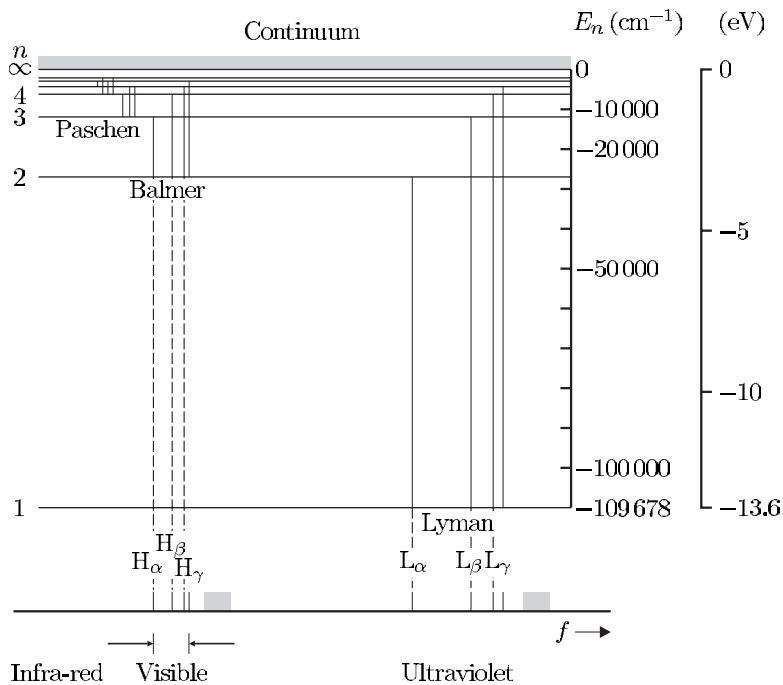


Fig. 1.1 The energy levels of the hydrogen atom. The transitions from higher shells $n' = 2, 3, 4, \dots$ down to the $n = 1$ shell give the Lyman series of spectral lines. The series of lines formed by transitions to other shells are: Balmer ($n = 2$), Paschen ($n = 3$), Brackett ($n = 4$) and Pfund ($n = 5$) (the last two are not labelled in the figure). Within each series the lines are denoted by Greek letters, e.g. L_α for $n = 2$ to $n = 1$ and H_β for $n = 4$ to $n = 2$.

1.3 Bohr's theory

In 1913, Bohr put forward a radical new model of the hydrogen atom using quantum mechanics. It was known from Rutherford's experiments that inside atoms there is a very small, dense nucleus with a positive charge. In the case of hydrogen this is a single proton with a single electron bound to it by the Coulomb force. Since the force is proportional to $1/r^2$, as for gravity, the atom can be considered in classical terms as resembling a miniature solar system with the electron orbiting around the proton, just like a planet going around the sun. However, quantum mechanics is important in small systems and only certain electron orbits are allowed. This can be deduced from the observation that hydrogen atoms emit light only at particular wavelengths corresponding to transitions between discrete energies. Bohr was able to explain the observed spectrum by introducing the then novel idea of quantisation that goes beyond any previous classical theory. He took the orbits that occur in classical mechanics and imposed quantisation rules onto them.

Bohr assumed that each electron orbits the nucleus in a circle, whose radius r is determined by the balance between centripetal acceleration and the Coulomb attraction towards the proton. For electrons of mass m_e and speed v this gives

$$\frac{m_e v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}. \quad (1.3)$$

In SI units the strength of the electrostatic interaction between two

⁵Older systems of units give more succinct equations without $4\pi\epsilon_0$; some of this neatness can be retained by keeping $e^2/4\pi\epsilon_0$ grouped together.

charges of magnitude e is characterised by the combination of constants $e^2/4\pi\epsilon_0$.⁵ This leads to the following relation between the angular frequency $\omega = v/r$ and the radius:

$$\omega^2 = \frac{e^2/4\pi\epsilon_0}{m_e r^3}. \quad (1.4)$$

This is equivalent to Kepler's laws for planetary orbits relating the square of the period $2\pi/\omega$ to the cube of the radius (as expected since all steps have been purely classical mechanics). The total energy of an electron in such an orbit is the sum of its kinetic and potential energies:

$$E = \frac{1}{2}m_e v^2 - \frac{e^2/4\pi\epsilon_0}{r}. \quad (1.5)$$

Using eqn 1.3 we find that the kinetic energy has a magnitude equal to half the potential energy (an example of the virial theorem). Taking into account the opposite signs of kinetic and potential energy, we find

$$E = -\frac{e^2/4\pi\epsilon_0}{2r}. \quad (1.6)$$

This total energy is negative because the electron is bound to the proton and energy must be supplied to remove it. To go further Bohr made the following assumption.

Assumption I There are certain allowed orbits for which the electron has a fixed energy. The electron loses energy only when it jumps between the allowed orbits and the atom emits this energy as light of a given wavelength.

That electrons in the allowed orbits do not radiate energy is contrary to classical electrodynamics—a charged particle in circular motion undergoes acceleration and hence radiates electromagnetic waves. Bohr's model does not explain why the electron does not radiate but simply takes this as an assumption that turns out to agree with the experimental data. We now need to determine which out of all the possible classical orbits are the allowed ones. There are various ways of doing this and we follow the standard method, used in many elementary texts, that assumes quantisation of the angular momentum in integral multiples of $\hbar$ (Planck's constant over 2π):

$$m_e v r = n \hbar, \quad (1.7)$$

where n is an integer. Combining this with eqn 1.3 gives the radii of the allowed orbits as

$$r = a_0 n^2, \quad (1.8)$$

where the Bohr radius a_0 is given by

$$a_0 = \frac{\hbar^2}{(e^2/4\pi\epsilon_0) m_e}. \quad (1.9)$$

This is the natural unit of length in atomic physics. Equations 1.6 and 1.8 combine to give the famous Bohr formula:

$$E = -\frac{e^2/4\pi\epsilon_0}{2a_0} \frac{1}{n^2}. \quad (1.10)$$

The positive integer n is called the **principal quantum number**.⁶

Bohr's formula predicts that in the transitions between these energy levels the atoms emit light with a wavenumber given by

$$\tilde{\nu} = R_{\infty} \left(\frac{1}{n^2} - \frac{1}{n'^2} \right). \quad (1.11)$$

This equation fits very closely to the observed spectrum of atomic hydrogen described by eqn 1.1. The Rydberg constant R_{∞} in eqn 1.11 is defined by

$$hcR_{\infty} = \frac{(e^2/4\pi\epsilon_0)^2 m_e}{2\hbar^2}. \quad (1.12)$$

The factor of hc multiplying the Rydberg constant is the conversion factor between energy and wavenumbers since the value of R_{∞} is given in units of m^{-1} (or cm^{-1} in commonly-used units). The measurement of the spectrum of atomic hydrogen using laser techniques has given an extremely accurate value for the Rydberg constant⁷ $R_{\infty} = 10\,973\,731.568\,525\,\text{m}^{-1}$. However, there is a subtle difference between the Rydberg constant calculated for an electron orbiting a fixed nucleus R_{∞} and the constant for real hydrogen atoms in eqn 1.1 (we originally wrote R without a subscript but more strictly we should specify that it is the constant for hydrogen R_{H}). The theoretical treatment above has assumed an infinitely massive nucleus, hence the subscript ∞ . In reality both the electron and proton move around the centre of mass of the system. For a nucleus of finite mass M the equations are modified by replacing the electron mass m_e by its reduced mass

$$m = \frac{m_e M}{m_e + M}. \quad (1.13)$$

For hydrogen

$$R_{\text{H}} = R_{\infty} \frac{M_{\text{p}}}{m_e + M_{\text{p}}} \simeq R_{\infty} \left(1 - \frac{m_e}{M_{\text{p}}} \right), \quad (1.14)$$

where the electron-to-proton mass ratio is $m_e/M_{\text{p}} \simeq 1/1836$. This reduced-mass correction is not the same for different isotopes of an element, e.g. hydrogen and deuterium. This leads to a small but readily observable difference in the frequency of the light emitted by the atoms of different isotopes; this is called the **isotope shift** (see Exercises 1.1 and 1.2).

1.4 Relativistic effects

Bohr's theory was a great breakthrough. It was such a radical change that the fundamental idea about the quantisation of the orbits was at first difficult for people to appreciate—they worried about how the electrons could know which orbits they were going into before they jumped. It was soon realised, however, that the assumption of circular orbits is

⁶The alert reader may wonder why this is true since we introduced n in connection with angular momentum in eqn 1.7, and (as shown later) electrons can have zero angular momentum. This arises from the simplification of Bohr's theory. Exercise 1.12 discusses a more satisfactory, but longer and subtler, derivation that is closer to Bohr's original papers. However, the important thing to remember from this introduction is not the formalism but the magnitude of the atomic energies and sizes.

⁷This is the 2002 CODATA recommended value. The currently accepted values of physical constants can be found on the web site of the National Institute of Science and Technology (NIST).

too much of an over-simplification. Sommerfeld produced a quantum mechanical theory of electrons in elliptical orbits that was consistent with special relativity. He introduced quantisation through a general rule that stated ‘the integral of the momentum associated with a coordinate around one period of the motion associated with that coordinate is an integral multiple of Planck’s constant’. This general method can be applied to any physical system where the classical motion is periodic. Applying this quantisation rule to momentum around a circular orbit gives the equivalent of eqn 1.7:⁸

$$m_e v \times 2\pi r = nh. \quad (1.15)$$

⁸This has a simple interpretation in terms of the de Broglie wavelength associated with an electron $\lambda_{dB} = h/m_e v$. The allowed orbits are those that have an integer multiple of de Broglie wavelengths around the circumference: $2\pi r = n\lambda_{dB}$, i.e. they are standing matter waves. Curiously, this idea has some resonance with modern ideas in string theory.

In addition to quantising the motion in the coordinate θ , Sommerfeld also considered quantisation of the radial degree of freedom r . He found that some of the elliptical orbits expected for a potential proportional to $1/r$ are also stationary states (some of the allowed orbits have a high eccentricity, more like those of comets than planets). Much effort was put into complicated schemes based on classical orbits with quantisation, and by incorporating special relativity this ‘old quantum theory’ could explain accurately the fine structure of spectral lines. The exact details of this work are now mainly of historical interest but it is worthwhile to make a simple estimate of relativistic effects. In special relativity a particle of rest mass m moving at speed v has an energy

$$E(v) = \gamma mc^2, \quad (1.16)$$

where the gamma factor is $\gamma = 1/\sqrt{1 - v^2/c^2}$. The kinetic energy of the moving particle is $\Delta E = E(v) - E(0) = (\gamma - 1)mc^2$. Thus relativistic effects produce a fractional change in energy:⁹

$$\frac{\Delta E}{E} \simeq \frac{v^2}{c^2}. \quad (1.17)$$

This leads to energy differences between the various elliptical orbits of the same gross energy because the speed varies in different ways around the elliptical orbits, e.g. for a circular orbit and a highly elliptical orbit of the same gross energy. From eqns 1.3 and 1.7 we find that the ratio of the speed in the orbit to the speed of light is

$$\frac{v}{c} = \frac{\alpha}{n}, \quad (1.18)$$

where the fine-structure constant α is given by

$$\alpha = \frac{e^2/4\pi\epsilon_0}{\hbar c}. \quad (1.19)$$

This fundamental constant plays an important role throughout atomic physics.¹⁰ Numerically its value is approximately $\alpha \simeq 1/137$ (see inside the back cover for a list of constants used in atomic physics). From eqn 1.17 we see that relativistic effects lead to energy differences of order α^2 times the gross energy. (This crude estimate neglects some

⁹We neglect a factor of $\frac{1}{2}$ in the binomial expansion of the expression for γ at low speeds, $v^2/c^2 \ll 1$.

¹⁰An electron in the Bohr orbit with $n = 1$ has speed αc . Hence it has linear momentum $m_e \alpha c$ and angular momentum $m_e \alpha c a_0 = \hbar$.

dependence on principal quantum number and Chapter 2 gives a more quantitative treatment of this fine structure.) It is not necessary to go into all the refinements of Sommerfeld's relativistic theory that gave the energy levels in hydrogen very precisely, by imposing quantisation rules on classical orbits, since ultimately a paradigm shift was necessary. Those ideas were superseded by the use of wavefunctions in the Schrödinger equation. The idea of elliptical orbits provides a connection with our intuition based on classical mechanics and we often retain some traces of this simple picture of electron orbits in our minds. However, for atoms with more than one electron, e.g. helium, classical models do not work and we must think in terms of wavefunctions.

1.5 Moseley and the atomic number

At the same time as Bohr was working on his model of the hydrogen atom, H. G. J. Moseley measured the X-ray spectra of many elements. Moseley established that the square root of the frequency of the emitted lines is proportional to the atomic number Z (that he defined as the position of the atom in the periodic table, starting counting at $Z = 1$ for hydrogen), i.e.

$$\sqrt{f} \propto Z. \quad (1.20)$$

Moseley's original plot is shown in Fig. 1.2. As we shall see, this equation is a considerable simplification of the actual situation but it was remarkably powerful at the time. By ordering the elements using Z rather than relative atomic mass, as was done previously, several inconsistencies in the periodic table were resolved. There were still gaps that were later filled by the discovery of new elements. In particular, for the rare-earth elements that have similar chemical properties and are therefore difficult to distinguish, it was said 'in an afternoon, Moseley could solve the problem that had baffled chemists for many decades and establish the true number of possible rare earths' (Segrè 1980). Moseley's observations can be explained by a relatively simple model for atoms that extends Bohr's model for hydrogen.¹¹

A natural way to extend Bohr's atomic model to heavier atoms is to suppose that the electrons fill up the allowed orbits starting from the bottom. Each energy level only has room for a certain number of electrons so they cannot all go into the lowest level and they arrange themselves in shells, labelled by the principal quantum number, around the nucleus. This shell structure arises because of the Pauli exclusion principle and the electron spin, but for now let us simply consider it as an empirical fact that the maximum number of electrons in the $n = 1$ shell is 2, the $n = 2$ shell has 8 and the $n = 3$ shell has 18, etc. For historical reasons, X-ray spectroscopists do not use the principal quantum number but label the shells by letters: K for $n = 1$, L for $n = 2$, M for $n = 3$ and so on alphabetically.¹² This concept of electronic shells explains the emission of X-rays from atoms in the following way. Moseley produced X-rays by bombarding samples of the given element with electrons that

¹¹Tragically, Henry Gwyn Jeffreys Moseley was killed when he was only 28 while fighting in the First World War (see the biography by Heilbron (1974)).

¹²The chemical properties of the elements depend on this electronic structure, e.g. the inert gases have full shells of electrons and these stable configurations are not willing to form chemical bonds. The explanation of the atomic structure underlying the periodic table is discussed further in Section 4.1. See also Atkins (1994) and Grant and Phillips (2001).

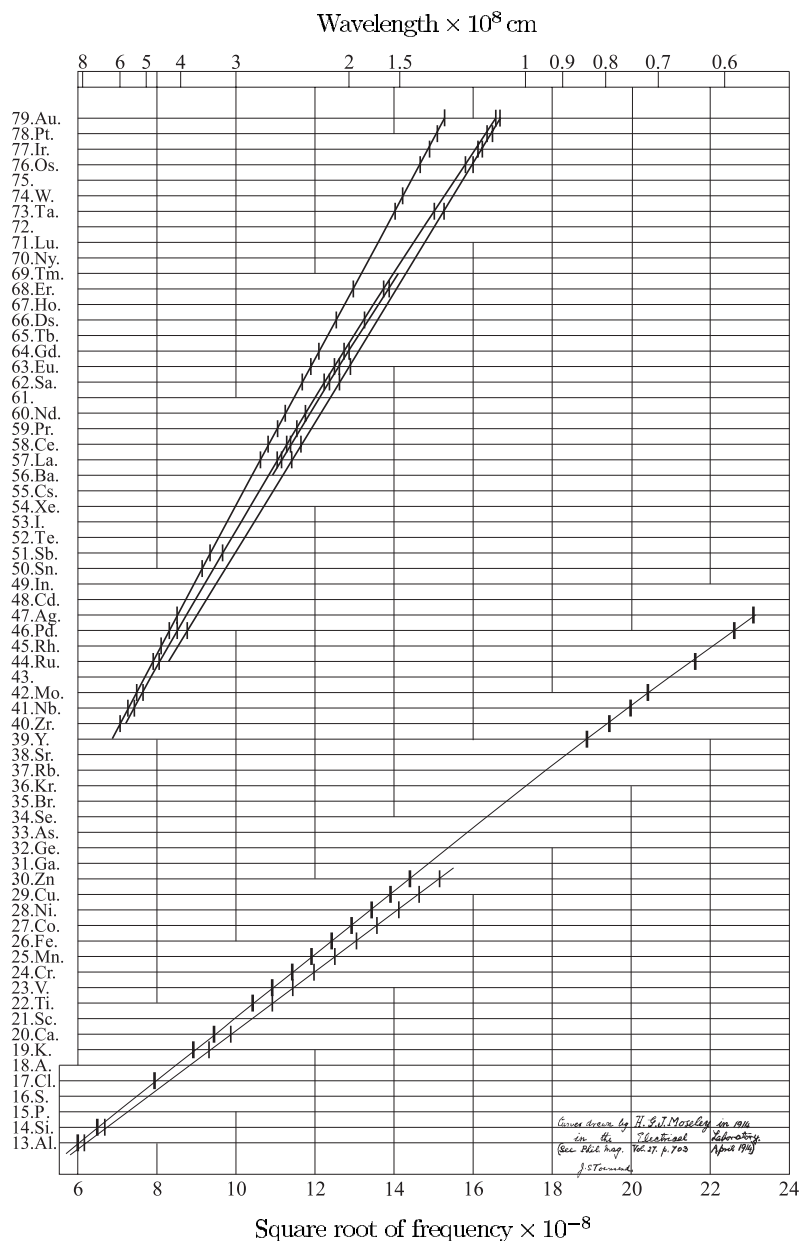


Fig. 1.2 Moseley's plot of the square root of the frequency of X-ray lines of elements against their atomic number. Moseley's work established the atomic number Z as a more fundamental quantity than the 'atomic weight' (now called relative atomic mass). Following modern convention the units of the horizontal scales would be $(10^8 \sqrt{\text{Hz}})$ at the bottom and (10^{-10} m) for the log scale at the top. (Archives of the Clarendon Laboratory, Oxford; also shown on the Oxford physics web site.)¹³

¹³The handwriting in the bottom right corner states that this diagram is the original for Moseley's famous paper in *Phil. Mag.*, **27**, 703 (1914).

had been accelerated to a high voltage in a vacuum tube. These fast electrons knock an electron out of an atom in the sample leaving a vacancy or hole in one of its shells. This allows an electron from a higher-lying shell to ‘fall down’ to fill this hole emitting radiation of a wavelength corresponding to the difference in energy between the shells.

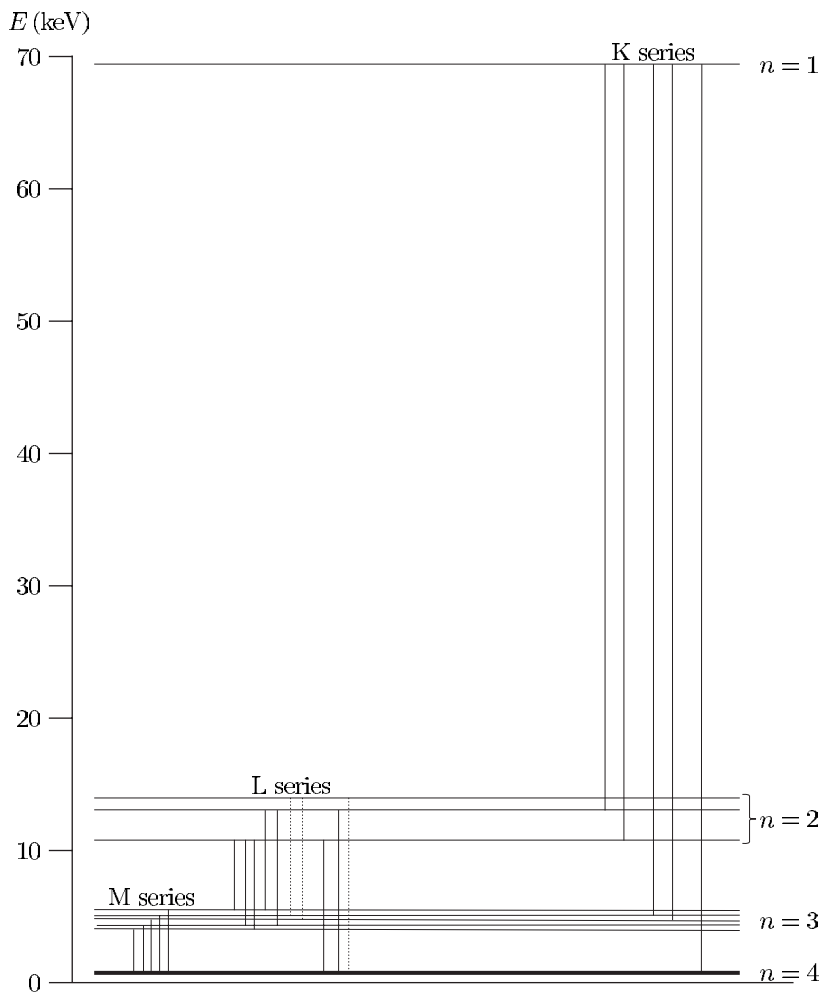
To explain Moseley’s observations quantitatively we need to modify the equations in Section 1.3, on Bohr’s theory, to account for the effect of a nucleus of charge greater than the $+1e$ of the proton. For a nuclear charge Ze we replace $e^2/4\pi\epsilon_0$ by $Ze^2/4\pi\epsilon_0$ in all the equations, resulting in a formula for the energies like that of Balmer but multiplied by a factor of Z^2 . This dependence on the square of the atomic number means that, for all but the lightest elements, transitions between low-lying shells lead to emission of radiation in the X-ray region of the spectrum. Scaling the Bohr theory result is accurate for hydrogenic ions, i.e. systems with one electron around a nucleus of charge Ze . In neutral atoms the other electrons (that do not jump) are not simply passive spectators but partly screen the nuclear charge; for a given X-ray line, say the K- to L-shell transition, a more accurate formula is

$$\frac{1}{\lambda} = R_{\infty} \left\{ \frac{(Z - \sigma_K)^2}{1^2} - \frac{(Z - \sigma_L)^2}{2^2} \right\}. \quad (1.21)$$

The screening factors σ_K and σ_L are not entirely independent of Z and the values of these screening factors for each shell vary slightly (see the exercises at the end of this chapter). For large atomic numbers this formula tends to eqn 1.20 (see Exercise 1.4). This simple approach does not explain why the screening factor for a shell can exceed the number of electrons inside that shell, e.g. $\sigma_K = 2$ for $Z = 74$ although only one electron remains in this shell when a hole is formed. This does not make sense in a classical model with electrons orbiting around a nucleus, but can be explained by atomic wavefunctions—an electron with a high principal quantum number (and little angular momentum) has a finite probability of being found at small radial distances.

The study of X-rays has developed into a whole field of its own within atomic physics, astrophysics and condensed matter, but there is only room to mention a few brief facts here. When an electron is removed from the K-shell the atom has an amount of energy equal to its binding energy, i.e. a positive amount of energy, and it is therefore usual to draw the diagram with the K-shell at the top, as in Fig. 1.3. These are the energy levels of the hole in the electron shells. This diagram shows why the creation of a hole in a low-lying shell leads to a succession of transitions as the hole works its way outwards through the shells. The hole (or equivalently the falling electron) can jump more than one shell at a time; each line in a series from a given shell is labelled using Greek letters (as in the series in hydrogen), e.g. K_{α} , K_{β} , ... The levels drawn in Fig. 1.3 have some sub-structure and this leads to transitions with slightly different wavelengths, as shown in Moseley’s plot. This is fine structure caused by relativistic effects that we considered for Sommerfeld’s theory; the substitution $e^2/4\pi\epsilon_0 \rightarrow Ze^2/4\pi\epsilon_0$, as above, (or

Fig. 1.3 The energy levels of the inner shells of the tungsten atom ($Z = 74$) and the transitions between them that give rise to X-rays. The level scheme has several important differences from that for the hydrogen atom (Fig. 1.1). Firstly, the energies are tens of keV, as compared to eV for $Z = 1$, because they scale as Z^2 (approximately). Secondly, the energy levels are plotted with $n = 1$ at the top because when an electron is removed from the K-shell the system has more energy than the neutral atom; energies are shown for an atom with a vacancy (missing electron) in the K-, L-, M- and N-shells. The atom emits X-ray radiation when an electron drops down from a higher shell to fill a vacancy in a lower shell—this process is equivalent to the vacancy, or hole, working its way outwards. This way of plotting the energies of the system shows clearly that the removal of an electron from the K-shell leads to a cascade of X-ray transitions, e.g. a transition between the $n = 1$ and 2 shells gives a line in the K-series which is followed by a line in another series (L-, M-, etc.). When the vacancy reaches the outermost shells of electrons that are only partially filled with valence electrons with binding energies of a few eV (the O- and P-shells in the case of tungsten), the transition energies become negligible compared to those between the inner shells. This level scheme is typical for electrons in a moderately heavy atom, i.e. one with filled K-, L-, M- and N-shells. (The lines of the L-series shown dotted are allowed X-ray transitions, but they do not occur following K_α emission.)



equivalently $\alpha \rightarrow Z\alpha$) shows that fine structure is of order $(Z\alpha)^2$ times the gross structure, which itself is proportional to Z^2 . Thus relativistic effects grow as Z^4 and become very significant for the inner electrons of heavy atoms, leading to the fine structure of the L- and M-shells seen in Fig. 1.3. This relativistic splitting of the shells explains why in Moseley's plot (Fig. 1.2) there are two closely-spaced curves for the K_α -line, and several curves for the L-series.

Nowadays much of the X-ray work in atomic physics is carried out using sources such as synchrotrons; these devices accelerate electrons by the techniques used in particle accelerators. A beam of high-energy electrons circulates in a ring and the circular motion causes the electrons to

radiate X-rays. Such a source can be used to obtain an X-ray absorption spectrum.¹⁴ There are many other applications of X-ray emission, e.g. as a diagnostic tool for the processes that occur in plasmas in fusion research and in astrophysical objects. Many interesting processes occur at ‘high energies’ in atomic physics but the emphasis in this book is mainly on lower energies.

1.6 Radiative decay

An electric dipole moment $-ex_0$ oscillating at angular frequency ω radiates a power¹⁵

$$P = \frac{e^2 x_0^2 \omega^4}{12\pi\epsilon_0 c^3}. \quad (1.22)$$

An electron in harmonic motion has a total energy¹⁶ of $E = m_e \omega^2 x_0^2 / 2$, where x_0 is the amplitude of the motion. This energy decreases at a rate equal to the power radiated:

$$\frac{dE}{dt} = -\frac{e^2 \omega^2}{6\pi\epsilon_0 m_e c^3} E = -\frac{E}{\tau}, \quad (1.23)$$

where the classical radiative lifetime τ is given by

$$\frac{1}{\tau} = \frac{e^2 \omega^2}{6\pi\epsilon_0 m_e c^3}. \quad (1.24)$$

For the transition in sodium at a wavelength of 589 nm (yellow light) this equation predicts a value of $\tau = 16 \text{ ns} \simeq 10^{-8} \text{ s}$. This is very close to the experimentally measured value and typical of allowed transitions that emit visible light. Atomic lifetimes, however, vary over a very wide range,¹⁷ e.g. for the Lyman- α transition (shown in Fig. 1.1) the upper level has a lifetime of only a few nanoseconds.^{18,19}

The classical value of the lifetime gives the fastest time in which the atom could decay on a given transition and this is often close to the observed lifetime for strong transitions. Atoms do not decay faster than a classical dipole radiating at the same wavelength, but they may decay more slowly (by many orders of magnitude in the case of forbidden transitions).²⁰

1.7 Einstein *A* and *B* coefficients

The development of the ideas of atomic structure was linked to experiments on the emission, and absorption, of radiation from atoms, e.g. X-rays or light. The emission of radiation was considered as something that just has to happen in order to carry away the energy when an electron jumps from one allowed orbit to another, but the mechanism was not explained.²¹ In one of his many strokes of genius Einstein devised a way of treating the phenomenon of spontaneous emission quantitatively,

¹⁴Absorption is easier to interpret than emission since only one of the terms in eqn 1.21 is important, e.g. $E_K = hcR_\infty(Z - \sigma_K)^2$.

¹⁵This total power equals the integral of the Poynting vector over a closed surface in the far-field of radiation from the dipole. This is calculated from the oscillating electric and magnetic fields in this region (see electromagnetism texts or Corny (2000)).

¹⁶The sum of the kinetic and potential energies.

¹⁷The classical lifetime scales as $1/\omega^2$. However, we will find that the quantum mechanical result is different (see Exercise 1.8).

¹⁸Higher-lying levels, e.g. $n = 30$, live for many microseconds (Gallagher 1994).

¹⁹Atoms can be excited up to configurations with high principal quantum numbers in laser experiments; such systems are called Rydberg atoms and have small intervals between their energy levels. As expected from the correspondence principle, these Rydberg atoms can be used in experiments that probe the interface between classical and quantum mechanics.

²⁰The ion-trapping techniques described in Chapter 12 can probe transitions with spontaneous decay rates less than 1 s^{-1} , using single ions confined by electric and magnetic fields—something that was only a ‘thought experiment’ for Bohr and the other founders of quantum theory. In particular, the effect of individual quantum jumps between atomic energy levels is observed. Radiative decay resembles radioactive decay in that individual atoms spontaneously emit a photon at a given time but taking the average over an ensemble of atoms gives exponential decay.

²¹A complete explanation of spontaneous emission requires quantum electrodynamics.

²²This treatment of the interaction of atoms with radiation forms the foundation for the theory of the laser, and is used whenever radiation interacts with matter (see Fox 2001). A historical account of Einstein's work and its profound implications can be found in Pais (1982).

²³The frequency dependence of the interaction is considered in Chapter 7.

²⁴The word **laser** is an acronym for light amplification by stimulated emission of radiation.

based on an intuitive understanding of the process.²²

Einstein considered atoms with two levels of energies, E_1 and E_2 , as shown in Fig. 1.4; each level may have more than one state and the number of states with the same energy is the **degeneracy** of that level represented by g_1 and g_2 . Einstein considered what happens to an atom interacting with radiation of energy density $\rho(\omega)$ per unit frequency interval. The radiation causes transitions from the lower to the upper level at a rate proportional to $\rho(\omega_{12})$, where the constant of proportionality is B_{12} . The atom interacts strongly only with that part of the distribution $\rho(\omega)$ with a frequency close to $\omega_{12} = (E_2 - E_1)/\hbar$, the atom's resonant frequency.²³ By symmetry it is also expected that the radiation will cause transitions from the upper to lower levels at a rate dependent on the energy density but with a constant of proportionality B_{21} (the subscripts are in a different order for emission as compared to absorption). This is a process of stimulated emission in which the radiation at angular frequency ω causes the atom to emit radiation of the same frequency. This increase in the amount of light at the incident frequency is fundamental to the operation of lasers.²⁴ The symmetry between up and down is broken by the process of spontaneous emission in which an atom falls down to the lower level, even when no external radiation is present. Einstein introduced the coefficient A_{21} to represent the rate of this process. Thus the rate equations for the populations of the levels, N_1 and N_2 , are

$$\frac{dN_2}{dt} = N_1 B_{12} \rho(\omega_{12}) - N_2 B_{21} \rho(\omega_{12}) - N_2 A_{21} \quad (1.25)$$

and

$$\frac{dN_1}{dt} = -\frac{dN_2}{dt}. \quad (1.26)$$

The first equation gives the rate of change of N_2 in terms of the absorption, stimulated emission and spontaneous emission, respectively. The second equation is a consequence of having only two levels so that atoms leaving level 2 must go into level 1; this is equivalent to a condition that $N_1 + N_2 = \text{constant}$. When $\rho(\omega) = 0$, and some atoms are initially in the upper level ($N_2(0) \neq 0$), the equations have a decaying exponential solution:

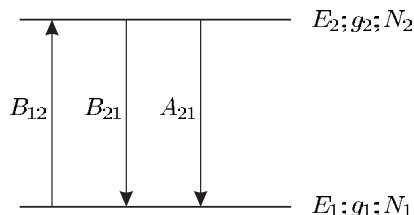
$$N_2(t) = N_2(0) \exp(-A_{21}t), \quad (1.27)$$

where the mean lifetime²⁵ is

$$\frac{1}{\tau} = A_{21}. \quad (1.28)$$

²⁵This lifetime was estimated by a classical argument in the previous section.

Fig. 1.4 The interaction of a two-level atom with radiation leads to stimulated transitions, in addition to the spontaneous decay of the upper level.



Einstein devised a clever argument to find the relationship between the A_{21} - and B -coefficients and this allows a complete treatment of atoms interacting with radiation. Einstein imagined what would happen to such an atom in a region of black-body radiation, e.g. inside a box whose surface acts as a black body. The energy density of the radiation $\rho(\omega) d\omega$ between angular frequency ω and $\omega + d\omega$ depends only on the temperature T of the emitting (and absorbing) surfaces of the box; this function is given by the Planck distribution law:²⁶

$$\rho(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{\exp(\hbar\omega/k_B T) - 1}. \quad (1.29)$$

Now we consider the level populations of an atom in this black-body radiation. At equilibrium the rates of change of N_1 and N_2 (in eqn 1.26) are both zero and from eqn 1.25 we find that

$$\rho(\omega_{12}) = \frac{A_{21}}{B_{21}} \frac{1}{(N_1/N_2)(B_{12}/B_{21}) - 1}. \quad (1.30)$$

At thermal equilibrium the population in each of the states within the levels are given by the Boltzmann factor (the population in each state equals that of the energy level divided by its degeneracy):

$$\frac{N_2}{g_2} = \frac{N_1}{g_1} \exp\left(-\frac{\hbar\omega}{k_B T}\right). \quad (1.31)$$

Combining the last three equations (1.29, 1.30 and 1.31) we find²⁷

$$A_{21} = \frac{\hbar\omega^3}{\pi^2 c^3} B_{21} \quad (1.32)$$

and

$$B_{12} = \frac{g_2}{g_1} B_{21}. \quad (1.33)$$

The Einstein coefficients are properties of the atom.²⁸ Therefore these relationships between them hold for any type of radiation, from narrow-bandwidth radiation from a laser to broadband light. Importantly, eqn 1.32 shows that strong absorption is associated with strong emission. Like many of the topics covered in this chapter, Einstein's treatment captured the essential features of the physics long before all the details of the quantum mechanics were fully understood.²⁹

1.8 The Zeeman effect

This introductory survey of early atomic physics must include Zeeman's important work on the effect of a magnetic field on atoms. The observation of what we now call the Zeeman effect and three other crucial experiments were carried out just at the end of the nineteenth century, and together these discoveries mark the watershed between classical and quantum physics.³⁰ Before describing Zeeman's work in detail, I shall

²⁶Planck was the first to consider radiation quantised into photons of energy $\hbar\omega$. See Pais (1986).

²⁷These equations hold for all T , so we can equate the parts that contain $\exp(\hbar\omega/k_B T)$ and the temperature-independent factors separately to obtain the two equations.

²⁸This is shown explicitly in Chapter 7 by a time-dependent perturbation theory calculation of B_{12} .

²⁹To excite a significant fraction of the population into the upper level of a visible transition would require black-body radiation with a temperature comparable to that of the sun, and this method is not generally used in practice—such transitions are easily excited in an electrical discharge where the electrons impart energy to the outermost electrons in an atom. (The voltage required to excite weakly-bound outer electrons is much less than for X-ray production.)

³⁰Pais (1986) and Segrè (1980) give historical accounts.

³¹This led to the measurement of the atomic X-ray spectra by Moseley described in Section 1.5.

³²The field of nuclear physics was later developed by Rutherford, and others, to show that atoms have a very small dense nucleus that contains almost all the atomic mass. For much of atomic physics it is sufficient to think of the nucleus as a positive charge $+Ze$ at the centre of the atoms. However, some understanding of the size, shape and magnetic moments of nuclei is necessary to explain the hyperfine structure and isotope shift (see Chapter 6).

³³This is the same force that Thomson used to deflect free electrons in a curved trajectory to measure e/m_e . Nowadays such cathode ray tubes are commonly used in classroom demonstrations.

briefly mention the other three great breakthroughs and their significance for atomic physics. Röntgen discovered mysterious X-rays emitted from discharges, and sparks, that could pass through matter and blacken photographic film.³¹ At about the same time, Becquerel's discovery of radioactivity opened up the whole field of nuclear physics.³² Another great breakthrough was J. J. Thomson's demonstration that cathode rays in electrical discharge tubes are charged particles whose charge-to-mass ratio does not depend on the gas in the discharge tube. At almost the same time, the observation of the Zeeman effect of a magnetic field showed that there are particles with the same charge-to-mass ratio in atoms (that we now call electrons). The idea that atoms contain electrons is very obvious now but at that time it was a crucial piece in the jigsaw of atomic structure that Bohr put together in his model. In addition to its historical significance, the Zeeman effect provides a very useful tool for examining the structure of atoms, as we shall see at several places in this book. Somewhat surprisingly, it is possible to explain this effect by a classical-mechanics line of reasoning (in certain special cases). An atom in a magnetic field can be modelled as a simple harmonic oscillator. The restoring force on the electron is the same for displacements in all directions and the oscillator has the same resonant frequency ω_0 for motion along the x -, y - and z -directions (when there is no magnetic field). In a magnetic field $\mathbf{B}$ the equation of motion for an electron with charge $-e$, position $\mathbf{r}$ and velocity $\mathbf{v} = \dot{\mathbf{r}}$ is

$$m_e \frac{d\mathbf{v}}{dt} = -m_e \omega_0^2 \mathbf{r} - e\mathbf{v} \times \mathbf{B}. \quad (1.34)$$

In addition to the restoring force (assumed to exist without further explanation), there is the Lorentz force that occurs for a charged particle moving through a magnetic field.³³ Taking the direction of the field to be the z -axis, $\mathbf{B} = B\hat{\mathbf{e}}_z$ leads to

$$\ddot{\mathbf{r}} + 2\Omega_L \dot{\mathbf{r}} \times \hat{\mathbf{e}}_z + \omega_0^2 \mathbf{r} = 0. \quad (1.35)$$

This contains the Larmor frequency

$$\Omega_L = \frac{eB}{2m_e}. \quad (1.36)$$

We use a matrix method to solve the equation and look for a solution in the form of a vector oscillating at ω :

$$\mathbf{r} = \text{Re} \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \exp(-i\omega t) \right\}. \quad (1.37)$$

Written in matrix form, eqn 1.35 reads

$$\begin{pmatrix} \omega_0^2 & -2i\omega\Omega_L & 0 \\ 2i\omega\Omega_L & \omega_0^2 & 0 \\ 0 & 0 & \omega_0^2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \omega^2 \begin{pmatrix} x \\ y \\ z \end{pmatrix}. \quad (1.38)$$

The eigenvalues ω^2 are found from the following determinant:

$$\begin{vmatrix} \omega_0^2 - \omega^2 & -2i\omega\Omega_L & 0 \\ 2i\omega\Omega_L & \omega_0^2 - \omega^2 & 0 \\ 0 & 0 & \omega_0^2 - \omega^2 \end{vmatrix} = 0. \quad (1.39)$$

This gives $\{\omega^4 - (2\omega_0^2 + 4\Omega_L^2)\omega^2 + \omega_0^4\}(\omega^2 - \omega_0^2) = 0$. The solution $\omega = \omega_0$ is obvious by inspection. The other two eigenvalues can be found exactly by solving the quadratic equation for ω^2 inside the curly brackets. For an optical transition we always have $\Omega_L \ll \omega_0$ so the approximate eigenfrequencies are $\omega \simeq \omega_0 \pm \Omega_L$. Substituting these values back into eqn 1.38 gives the eigenvectors corresponding to $\omega = \omega_0 - \Omega_L$, ω_0 and $\omega_0 + \Omega_L$, respectively, as

$$\mathbf{r} = \begin{pmatrix} \cos(\omega_0 - \Omega_L)t \\ -\sin(\omega_0 - \Omega_L)t \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ \cos\omega_0 t \end{pmatrix}$$

and $\begin{pmatrix} \cos(\omega_0 + \Omega_L)t \\ \sin(\omega_0 + \Omega_L)t \\ 0 \end{pmatrix}$

The magnetic field does not affect motion along the z -axis and the angular frequency of the oscillation remains ω_0 . Interaction with the magnetic field causes the motions in the x - and y -directions to be coupled together (by the off-diagonal elements $\pm 2i\omega\Omega_L$ of the matrix in eqn 1.38).³⁴ The result is two circular motions in opposite directions in the xy -plane, as illustrated in Fig. 1.5. These circular motions have frequencies shifted up, or down, from ω_0 by the Larmor frequency. Thus the action of the external field splits the original oscillation at a single frequency (actually three independent oscillations all with the same frequency, ω_0) into three separate frequencies. An oscillating electron acts as a classical dipole that radiates electromagnetic waves and Zeeman observed the frequency splitting Ω_L in the light emitted by the atom.

This classical model of the Zeeman effect explains the polarization of the light, as well as the splitting of the lines into three components. The calculation of the polarization of the radiation at each of the three different frequencies for a general direction of observation is straightforward using vectors;³⁵ however, only the particular cases where the radiation propagates parallel and perpendicular to the magnetic field are considered here, i.e. the longitudinal and transverse directions of observation, respectively. An electron oscillating parallel to $\mathbf{B}$ radiates an electromagnetic wave with linear polarization and angular frequency ω_0 . This π -component of the line is observed in all directions *except* along the magnetic field;³⁶ in the special case of transverse observation (i.e. in the xy -plane) the polarization of the π -component lies along $\hat{\mathbf{e}}_z$. The circular motion of the oscillating electron in the xy -plane at angular frequencies $\omega_0 + \Omega_L$ and $\omega_0 - \Omega_L$ produces radiation at these frequencies. Looking transversely, this circular motion is seen edge-on so that it looks like linear sinusoidal motion, e.g. for observation along

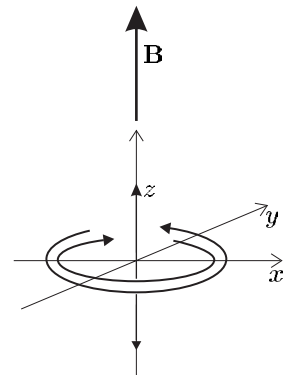


Fig. 1.5 A simple model of an atom as an electron that undergoes simple harmonic motion explains the features of the normal Zeeman effect of a magnetic field (along the z -axis). The three eigenvectors of the motion are: $\hat{\mathbf{e}}_z \cos \omega_0 t$ and $\cos(\{\omega_0 \pm \Omega_L\}t) \hat{\mathbf{e}}_x \pm \sin(\{\omega_0 \pm \Omega_L\}t) \hat{\mathbf{e}}_y$.

³⁴The matrix does not have off-diagonal elements in the last column or bottom row, so the x - and y -components are not coupled to the z -component, and the problem effectively reduces to solving a 2×2 matrix.

³⁵Some further details are given in Section 2.2 and in Woodgate (1980).

³⁶An oscillating electric dipole proportional to $\hat{\mathbf{e}}_z \cos \omega_0 t$ does not radiate along the z -axis—observation along this direction gives a view along the axis of the dipole so that effectively the motion of the electron cannot be seen.

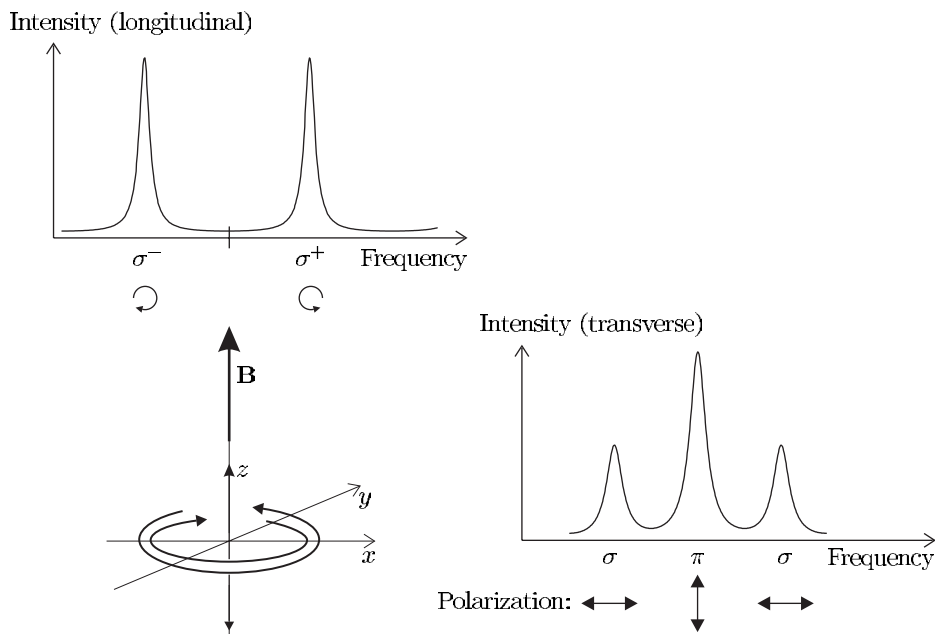


Fig. 1.6 For the normal Zeeman effect a simple model of an atom (as in Fig. 1.5) explains the frequency of the light emitted and its polarization (indicated by the arrows for the cases of transverse and longitudinal observation).

the x -axis only the y -component is seen, and the radiation is linearly polarized perpendicular to the magnetic field—see Fig. 1.6. These are called the σ -components and, in contrast to the π -component, they are also seen in longitudinal observation—looking along the z -axis one sees the electron's circular motion and hence light that has circular polarization. Looking in the opposite direction to the magnetic field (from the positive z -direction, or $\theta = 0$ in polar coordinates) the circular motion in the anticlockwise direction is associated with the frequency $\omega_0 + \Omega_L$.³⁷ In addition to showing that atoms contain electrons by measuring the magnitude of the charge-to-mass ratio e/m_e , Zeeman also deduced the sign of the charge by considering the polarization of the emitted light. If the sign of the charge was not negative, as we assumed from the start, light at $\omega_0 + \Omega_L$ would have the opposite handedness—from this Zeeman could deduce the sign of the electron's charge.

For situations that only involve orbital angular momentum (and no spin) the predictions of this classical model correspond exactly to those of quantum mechanics (including the correct polarizations), and the intuition gained from this model gives useful guidance in more complicated cases. Another reason for studying the classical treatment of the Zeeman effect is that it furnishes an example of degenerate perturbation theory in classical mechanics. We shall encounter degenerate perturbation theory in quantum mechanics in several places in this book and an understanding of the analogous procedure in classical mechanics is very helpful.

³⁷This is left-circularly-polarized light (Corney 2000).

1.8.1 Experimental observation of the Zeeman effect

Figure 1.7(a) shows an apparatus suitable for the experimental observation of the Zeeman effect and Fig. 1.7(b–e) shows some typical experimental traces. A low-pressure discharge lamp that contains the atom to be studied (e.g. helium or cadmium) is placed between the pole pieces of an electromagnet capable of producing fields of up to about 1 T. In the arrangement shown, a lens collects light emitted perpendicular to the field (transverse observation) and sends it through a Fabry–Perot étalon. The operation of such étalons is described in detail by Brooker (2003), and only a brief outline of the principle of operation is given here.

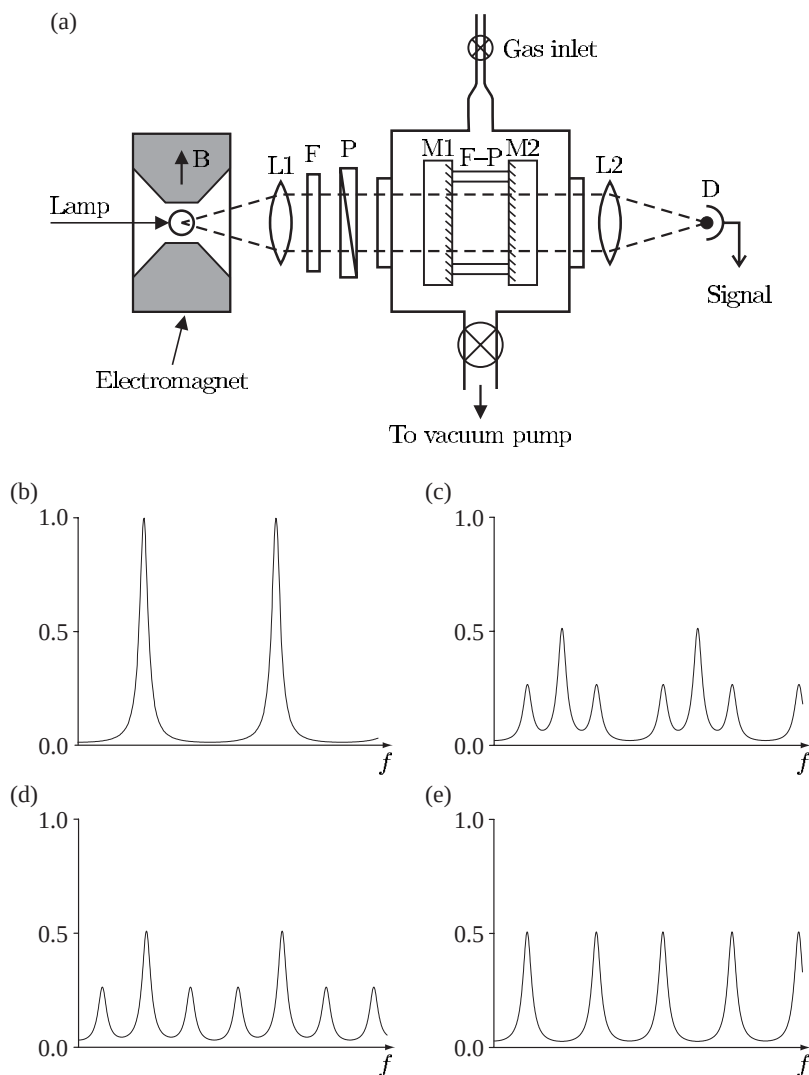


Fig. 1.7 (a) An apparatus suitable for the observation of the Zeeman effect. The light emitted from a discharge lamp, between the pole pieces of the electromagnet, passes through a narrow-band filter and a Fabry–Perot étalon. Key: L1, L2 are lenses; F—filter; P—polarizer to discriminate between π - and σ -polarizations (optional); Fabry–Perot étalon made of a rigid spacer between two highly-reflecting mirrors (M1 and M2); D—detector. Other details can be found in Brooker (2003). A suitable procedure is to (partially) evacuate the étalon chamber and then allow air (or a gas with a higher refractive index such as carbon dioxide) to leak in through a constant-flow-rate valve to give a smooth linear scan. Plots (b) to (e) show the intensity I of light transmitted through the Fabry–Perot étalon. (b) A scan over two free-spectral ranges with no magnetic field. Both (c) and (d) show a Zeeman pattern observed perpendicular to the applied field; the spacing between the π - and σ -components in these scans is one-quarter and one-third of the free-spectral range, respectively—the magnetic field in scan (c) is weaker than in (d). (e) In longitudinal observation only the σ -components are observed—this scan is for the same field as in (c) and the σ -components have the same position in both traces.

- Light from the lamp is collected by a lens and directed on to an interference filter that transmits only a narrow band of wavelengths corresponding to a single spectral line.
- The étalon produces an interference pattern that has the form of concentric rings. These rings are observed on a screen in the focal plane of the lens placed after the étalon. A small hole in the screen is positioned at the centre of the pattern so that light in the region of the central fringe falls on a detector, e.g. a photodiode. (Alternatively, the lens and screen can be replaced by a camera that records the ring pattern on film.)
- The effective optical path length between the two flat highly-reflecting mirrors is altered by changing the pressure of the air in the chamber; this scans the étalon over several free-spectral ranges while the intensity of the interference fringes is recorded to give traces as in Fig. 1.7(b–e).

1.9 Summary of atomic units

This chapter has used classical mechanics and elementary quantum ideas to introduce the important scales in atomic physics: the unit of length a_0 and a unit of energy hcR_∞ . The natural unit of energy is $e^2/4\pi\epsilon_0 a_0$ and this unit is called a hartree.³⁸ This book, however, expresses energy in terms of the energy equivalent to the Rydberg constant, 13.6 eV; this equals the binding energy in the first Bohr orbit of hydrogen, or 1/2 a hartree. These quantities have the following values:

$$a_0 = \frac{\hbar^2}{(e^2/4\pi\epsilon_0) m_e} = 5.29 \times 10^{-11} \text{ m}, \quad (1.40)$$

$$hcR_\infty = \frac{m_e (e^2/4\pi\epsilon_0)^2}{2\hbar^2} = 13.6 \text{ eV}. \quad (1.41)$$

The use of these atomic units makes the calculation of other quantities simple, e.g. the electric field in a hydrogen atom at radius $r = a_0$ equals $e/(4\pi\epsilon_0 a_0^2)$. This corresponds to a potential difference of 27.2 V over a distance of a_0 , or a field of $5 \times 10^{11} \text{ V m}^{-1}$.

Relativistic effects depend on the dimensionless fine-structure constant α :

$$\alpha = \frac{(e^2/4\pi\epsilon_0)}{\hbar c} \simeq \frac{1}{137}. \quad (1.42)$$

The Zeeman effect of a magnetic field on atoms leads to a frequency shift of Ω_L in eqn 1.36.³⁹ In practical units the size of this frequency shift is

$$\frac{\Omega_L}{2\pi B} = \frac{e}{4\pi m_e} = 14 \text{ GHz T}^{-1}. \quad (1.43)$$

Equating the magnetic energy $\hbar\Omega_L$ with $\mu_B B$, the magnitude of the energy for a magnetic moment μ_B in a magnetic flux density B , shows that the unit of atomic magnetic moment is the Bohr magneton

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \text{ J T}^{-1}. \quad (1.44)$$

³⁸It equals the potential energy of the electron in the first Bohr orbit.

³⁹This Larmor frequency equals the splitting between the π - and σ -components in the normal Zeeman effect.

This magnetic moment depends on the properties of the unpaired electron (or electrons) in the atom, and has a similar magnitude for all atoms. In contrast, other atomic properties scale rapidly with the nuclear charge; hydrogenic systems have energies proportional to Z^2 , and the same reasoning shows that their size is proportional to $1/Z$ (see eqns 1.40 and 1.41). For example, hydrogenic uranium U^{+91} has been produced in accelerators by stripping 91 electrons off a uranium atom to leave a single electron that has a binding energy of $92^2 \times 13.6 \text{ eV} = 115 \text{ keV}$ (for $n = 1$) and an orbit of radius $a_0/92 = 5.75 \times 10^{-13} \text{ m} \equiv 575 \text{ fm}$. The transitions between the lowest energy levels of this system have short wavelengths in the X-ray region.⁴⁰

The reader might think that it would be a good idea to use the same units across the whole of atomic physics. In practice, however, the units reflect the actual experimental techniques used in a particular region of the spectrum, e.g. radio-frequency, or microwave synthesisers, are calibrated in Hz (kHz, MHz and GHz); the equation for the angle of diffraction from a grating is expressed in terms of a wavelength; and for X-rays produced by tubes in which electrons are accelerated by high voltages it is natural to use keV.⁴¹ A table of useful conversion factors is given inside the back cover.

The survey of classical ideas in this chapter gives a historical perspective on the origins of atomic physics but it is not necessary, or indeed in some cases downright confusing, to go through a detailed classical treatment—the physics at the scale of atomic systems can only properly be described by wave mechanics and this is the approach used in the following chapters.⁴²

⁴⁰Energies can be expressed in terms of the rest mass energy of the electron $m_e c^2 = 0.511 \text{ MeV}$. The gross energy is $(Z\alpha)^2 \frac{1}{2} m_e c^2$ and the fine structure is of order $(Z\alpha)^4 \frac{1}{2} m_e c^2$.

⁴¹Laser techniques can measure transition frequencies of around 10^{15} Hz directly as a frequency to determine a precise value of the Rydberg constant, and there are no definite rules for whether a transition should be specified by its energy, wavelength or frequency.

⁴²X-ray spectra are not discussed again in this book and further details can be found in Kuhn (1969) and other atomic physics texts.

Exercises

(1.1) *Isotope shift*

The deuteron has approximately twice the mass of the proton. Calculate the difference in the wavelength of the Balmer- α line in hydrogen and deuterium.

(1.2) *The energy levels of one-electron atoms*

The table gives the wavelength⁴³ of lines observed in the spectrum of atomic hydrogen and singly-ionized helium. Explain as fully as possible the similarities and differences between the two spectra.

H (nm)	He ⁺ (nm)
656.28	656.01
486.13	541.16
434.05	485.93
410.17	454.16
	433.87
	419.99
	410.00

⁴³These are the wavelengths in air with a refractive index of 1.0003 in the visible region.

(1.3) *Relativistic effects*

Evaluate the magnitude of relativistic effects in the $n = 2$ level of hydrogen. What is the resolving power $\lambda/(\Delta\lambda)_{\min}$ of an instrument that could observe these effects in the Balmer- α line?

(1.4) *X-rays*

Show that eqn 1.21 approximates to eqn 1.20 when the atomic number Z is much greater than the screening factors.

(1.5) *X-rays*

It is suspected that manganese ($Z = 25$) is very poorly mixed with iron ($Z = 26$) in a block of alloy. Predict the energies of the K-absorption edges of these elements and determine an X-ray photon energy that would give good contrast (between regions of different concentrations) in an X-ray of the block.

(1.6) *X-ray experiments*

Sketch an apparatus suitable for X-ray spectroscopy of elements, e.g. Moseley's experiment. Describe the principle of its operation and the method of measuring the energy, or wavelength, of X-rays.

(1.7) *Fine structure in X-ray transitions*

Estimate the magnitude of the relativistic effects in the L-shell of lead ($Z = 82$) in keV. Also express your answer as a fraction of the K_α transition.

(1.8) *Radiative lifetime*

For an electron in a circular orbit of radius r the electric dipole moment has a magnitude of $D = -er$ and radiates energy at a rate given by eqn 1.22. Find the time taken to lose an energy of $\hbar\omega$.

Use your expression to estimate the transition rate for the $n = 3$ to $n = 2$ transition in hydrogen that emits light of wavelength 656 nm.

Comment. This method gives $1/\tau \propto (er)^2 \omega^3$, which corresponds closely to the quantum mechanical result in eqn 7.23.

(1.9) *Black-body radiation*

Two-level atoms with a transition at wavelength $\lambda = 600$ nm, between the levels with degeneracies $g_1 = 1$ and $g_2 = 3$, are immersed in black-body radiation. The fraction in the excited state is 0.1. What is the temperature of the black body and the energy density per unit frequency interval $\rho(\omega_{12})$ of the radiation at the transition frequency?

(1.10) *Zeeman effect*

What is the magnitude of the Zeeman shift for an atom in (a) the Earth's magnetic field, and (b) a

magnetic flux density of 1 T? Express your answers in both MHz, and as a fraction of the transition frequency $\Delta f/f$ for a spectral line in the visible.

(1.11) *Relative intensities in the Zeeman effect*

Without an external field, an atom has no preferred direction and the choice of quantisation axis is arbitrary. In these circumstances the light emitted cannot be polarized (since this would establish a preferred orientation). As a magnetic field is gradually turned on we do not expect the intensities of the different Zeeman components to change discontinuously because the field has little effect on transition rates. This physical argument implies that oppositely-polarized components emitted along the direction of the field must have equal intensities, i.e. $I_{\sigma+} = I_{\sigma-}$ (notation defined in Fig. 1.6). What can you deduce about

(a) the relative intensities of the components emitted perpendicularly to the field?

(b) the ratio of the total intensities of light emitted along and perpendicular to the field?

(1.12) *Bohr theory and the correspondence principle*

This exercise gives an alternative approach to the theory of the hydrogen atom presented in Section 1.3 that is close to the spirit of Bohr's original papers. It is somewhat more subtle than that usually given in elementary textbooks and illustrates Bohr's great intuition. Rather than the *ad hoc* assumption that angular momentum is an integral multiple of $\hbar$ (in eqn 1.7), Bohr used the correspondence principle. This principle relates the behaviour of a system according to the known laws of classical mechanics and its quantum properties.

Assumption II The correspondence principle states that in the limit of large quantum numbers a quantum system tends to the same limit as the corresponding classical system.

Bohr formulated this principle in the early days of quantum theory. To apply this principle to hydrogen we first calculate the energy gap between adjacent electron orbits of radii r and r' . For large radii, the change $\Delta r = r' - r \ll r$.

(a) Show that the angular frequency $\omega = \Delta E/\hbar$ of radiation emitted when an electron makes a quantum jump between these levels is

$$\omega \simeq \frac{e^2/4\pi\epsilon_0}{2\hbar} \frac{\Delta r}{r^2}.$$

(b) An electron moving in a circle of radius r acts as an electric dipole radiating energy at the

orbital frequency ω given by eqn 1.4. Verify that this equation follows from eqn 1.3.

- (c) In the limit of large quantum numbers, the quantum mechanical and classical expressions give the same frequency ω . Show that equating the expressions in the previous parts yields $\Delta r = 2(a_0 r)^{1/2}$.
- (d) The difference in the radii between two adjacent orbits can be expressed as a difference equation.⁴⁴ In this case $\Delta n = 1$ and

$$\frac{\Delta r}{\Delta n} \propto r^{1/2}. \quad (1.45)$$

This equation can be solved by assuming that the radius varies as some power x of the quantum number n , e.g. if one orbit is labelled by an integer n and the next by $n + 1$, then $r = an^x$ and $r' = a(n + 1)^x$. Show that $\Delta r = axn^{x-1} \propto n^{x/2}$. Determine the power x and the constant a .

Comment. We have found eqn 1.8 from the correspondence principle without considering angular momentum. The allowed energy levels are easily found from this equation as in Section 1.3. The remarkable feature is that, although the form of the equation was derived for high values of the principal quantum number, the result works down to $n = 1$.

(1.13) *Rydberg atoms*

- (a) Show that the energy of the transitions between two shells with principal quantum numbers n and $n' = n + 1$ is proportional to $1/n^3$ for large n .
- (b) Calculate the frequency of the transition between the $n' = 51$ and $n = 50$ shells of a neutral atom.
- (c) What is the size of an atom in these *Rydberg states*? Express your answer both in atomic units and in metres.

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.

⁴⁴A difference equation is akin to a differential equation but without letting the differences become infinitesimal.